Quiz 1

ETL — the three stages, and the order

Covers	Extract, transform, load — and what each stage guarantees
Tasks	T1, T2 and T3 of the ETL workbook
Marks	**44** (of the 95 workbook marks for this dataset)
CLO	CLO3 · Competence
Component	Hands-on Exercises (20% of the course)
Work	**Individual**
Tool needed	**None.** A spreadsheet and the three source files
Submit	A **GitHub repository link** on the LMS — folder `quiz1/`
Due	As posted on the LMS

What this quiz is

The ETL workbook has seven tasks. **Quiz 1 is the first three** — the paper design, done before anybody opens a tool. Quiz 2 is the other four, and it needs a working pipeline.

The division is not arbitrary. T1 to T3 are fifty-five minutes of reading, counting and arguing, and none of it requires Power BI, Power Query or Python. You need the three dirty files, a spreadsheet, and the discipline to not fix anything while you read.

> **Do T1 before T3.** You cannot order a pipeline against a mess you have not counted. The workbook is ordered this way deliberately.

The case

A commercial bank pulls a regional currency-dealing log every night at 02:00 — one file per region, written by a system nobody at the bank controls. By 08:00 the treasury desk needs one number on screen.

Last night the file came in with eleven empty rows, branch codes typed as `LHR-04`, `lhr-04` and `Lhr 04`, and amounts carrying thousands separators and a stray en-dash where a minus sign belonged. **The load did not fail. It succeeded, partly.** At 08:00 the desk read a position short by one region and spent the morning trading against it. Nobody was told, because nothing crashed.

Then the same story in your own data. You have already opened three PakFreight files, fixed them by hand, and pasted the result into one sheet — a correct table and zero reusable work. You have also wired an empty star and left it waiting.

Now those two halves meet: you build the machine that loads the star. The deliverable is **not the table** — the table has already been produced by hand once. It is the **seventeen steps, in order, with their names on them**, and the test of it is that tomorrow night it runs without you.

The data you need

Everything for this quiz is in the `data/` folder distributed with it.

File	What it is
`data/PakFreight_ETLLab_STUDENT.xlsx`	**Start here.** Sheets T1, T2 and T3 are this quiz. Cream cells are yours to fill

File	What it is
`data/source/marine_shipments.csv`	Dates as `dd/MM/yyyy` **text**, status UPPER CASE
`data/source/rail_shipments.xlsx`	Two title rows above the header, `RL–` prefix stripped, weight in **TONNES**
`data/source/road_shipments.csv`	One blank line, one `TOTAL` footer, lower-case status
`data/dims/*.csv`	The five dimensions of the star
`data/alias_lookup.csv`	The 23-row alias map, to **check** your own against

Thirty-seven lines across three files, ten defects. That is the whole input for T1.

> The refresh-test folder is **not** in this pack. It belongs to Quiz 2 — it is there to test a pipeline, not to help you build one.
>
> No answer key is distributed with this quiz, and none exists in any repository you have access to.

The rules

These are the rules of the exercise. Most of them bear directly on this quiz.

- **Extract corrects nothing.** Thirty-seven rows arrive and thirty-seven rows land, blank line and `TOTAL` footer included. Fix anything on the way in and you have no untouched copy to prove the source against, and no way to tell a source problem from your own.
- **Shape, then text, then type, then keys.** Four words in that order. Anybody who gets stuck will have tried to look up a key before trimming the value it is looking up.
- **Order is a mechanism, not a style.** Three orderings are forbidden and everything else is taste. Know which three, and know the number that each one costs you when you break it.
- **Name every step.** `RemovedTotalRow` tells the next person what `FilteredRows1` never will. It takes two seconds, it is the only documentation the pipeline will ever have, and it is marked.
- **Nothing is dropped silently.** A row you cannot resolve goes to the Unknown member at `-1` and is **counted**. Deleting it is an answer, and you have to defend it in writing.

Type only in cream cells. Grey cells are given to you: the defect list, the step list, the alias map.

Every number in the workbook came out of `data/source`. So can yours. Open the files and count — nothing here has to be taken on trust.

T1 · Defect census · 12 marks

Ten defects across three files totalling thirty-seven lines. For each one: which of the **four transform classes** fixes it (Structural, String, Type & Unit, Conform), **which step** clears it, and whether ignoring it fails **loudly or quietly**.

Use these definitions literally:

- **Loud** — the refresh errors, or a step fails outright, and somebody is told.
- **Quiet** — the load succeeds and a published number is wrong.

You produce: a class, a step and a verdict for all ten rows, plus one line each on what breaks if you ignore it.

Self-check (in the sheet): exactly **1** of the ten fails loudly, **7** fail quietly, and **1** does both.

> If your Loud count is 5 or 6 you are marking "I would notice that" as loud. The test is not whether a careful person would spot it — it is whether the **pipeline tells somebody**. Exactly one does. The other nine lie with a straight face.

One of the ten rows is not a defect at all. Find it and say why it is on the list. **4 of the 12 marks** are for that answer and for one on why defects 8 and 9 — the `TOTAL` footer and the repeated consignment — are judgements rather than errors.

T2 · Stage ownership · 12 marks

Twelve activities. For each: **Extract**, **Transform**, **Load** — or **Violation**, meaning one stage is doing another's job, or a job no stage may do at all.

The value of the three-word name is that it **divides the blame**: when a pipeline breaks you should be able to say which of the three stages broke. A stage doing another stage's job makes that impossible, which is why the three "must nots" are the part that gets skipped:

- Extract must not clean.
- Transform must not write to the source.
- Load must not decide anything.

You produce: a verdict and a reason for all twelve. One mark for the verdict, one for the reason.

Self-check (in the sheet): 4 of the twelve are violations, and **Transform** owns more of the rest than either other stage — which is not an accident of how the list was written.

> The four violations are there to be found by argument, not by elimination. The other eight are real work that a real stage really owns.

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4 of the 12 marks are for one paragraph on why staging is never user-facing, naming the Power Query setting that enforces it.

T3 · Order the seventeen steps · 20 marks

The seventeen steps are listed **alphabetically** in the sheet, not in run order. Put them in an order that works, and for each one say which stage owns it and whether it runs **per source** or **once** on the combined table.

You produce: a run-order number 1–17 used exactly once each, a stage, a cadence, the defects that step clears, and one line on why it sits where you put it.

Self-check (in the sheet): 3 extract + 10 transform + 4 load, and **10 run per source** while **7 run once**.

> Getting the cadence column wrong is how you end up appending three already-deduplicated tables and still finding a duplicate. Ask of every step: *does this need information that only exists before the append?*

The three constraints · 9 of the 20 marks

Three orderings are not negotiable. Name what each one costs you, **as a number**:

	Constraint	What breaking it costs
1	Rename before append	a fact table of **20 columns**
2	Trim before deduplicate	a duplicate that survives the dedup step
3	Convert before append	**283,695.25 kg** instead of 578,650 kg

And the question that separates a rule from a habit · 4 of the 20 marks

Run the dedup step **before** the trim step on tonight's road file and you **still get 34 rows**. So why is "trim before dedup" a constraint and not a preference?

> This is the hardest question in the workbook and it is worth answering properly. Look at the two `RD-505` rows before you answer it.
>
> The two `RD-505` rows are **byte-identical**, so both dedup rules give 34 rows and 578,650 kg. Noticing that is **correct and earns marks** — and then the rule still holds, for reasons the task asks you to find.

Marking guide

Task	What is marked	Marks
T1	Class, step and loud/quiet for all ten defects (8) + the row that is not a defect, and defects 8 and 9 as judgements (4)	**12**
T2	Verdict and reason for all twelve activities (8) + the staging-is-not-user-facing paragraph, naming the setting (4)	**12**
T3	Run order 1–17, stage and cadence, with a line of justification each (7) + the three constraints, each costed as a number (9) + trim-before-dedup as a rule and not a preference (4)	**20**
	Quiz 1 total	**44**

A verdict with no reason scores half. A constraint named without its number scores nothing — *"the totals come out wrong"* is not an answer when the task asks for **283,695.25 kg**.

How to submit

You submit a link, not a file. One repository for the whole semester, one folder per assessment.

1. Create one public GitHub repository for this course, if you have not already:

```
DW-<your roll number>
```

One repository for every assessment in this course. Do **not** create a new repository per quiz.

2. Create this folder inside it, exactly:

```
quiz1/
```

3. Put these three files in that folder:

File	What it is
`Quiz1_<roll number>.xlsx`	The workbook with **sheets T1, T2 and T3** filled. Later sheets may be empty — they are Quiz 2
`answers.md`	The written parts, in your own words: the row that is not a defect; defects 8 and 9; the staging paragraph; the three constraints with their numbers; trim-before-dedup
`README.md`	Your name, roll number, and which spreadsheet you used

4. Commit and push:

```
git add quiz1
git commit -m "Quiz 1"
git push
```

5. On the LMS, submit the repository link only:

```
https://github.com/<your-username>/DW-<your roll number>
```

Paste it into the text box. **No attachments.** If your repository is private, the submission cannot be marked — make it public, or add the instructor as a collaborator and say so in the LMS comment.

> A link to a repository with an empty `quiz1/` folder is a zero. Check your own link in a logged-out browser window before you submit it.

Definition of done

- ☐ Sheets T1, T2 and T3 complete, every cream cell filled
- ☐ T1: ten classes, ten steps, ten verdicts — **1 loud, 7 quiet, 1 both**
- ☐ T1: the row that is not a defect, named, with the reason
- ☐ T2: twelve verdicts with twelve reasons — **4 violations**
- ☐ T2: the staging paragraph, naming the Power Query setting

☐ T3: 1–17 each used exactly once; **3 extract + 10 transform + 4 load**; **10 per source + 7 once**
☐ T3: all three constraints costed **as numbers**
☐ T3: the trim-before-dedup answer, written against the two `RD-505` rows
☐ `answers.md` and `README.md` present
☐ Pushed, public, and the link opens for a stranger

References

- **Kimball & Ross (2013)**, *The Data Warehouse Toolkit*, 3rd ed. — Ch. 19 on the ETL subsystems, of which this exercise builds about nine; Ch. 20 on staging and dependency order.
- **Kimball & Caserta (2004)**, *The Data Warehouse ETL Toolkit* — Chapters 4 and 5 on cleaning and conforming, the source of the defect-class argument.
- **Inmon (2005)**, *Building the Data Warehouse*, 4th ed. — the normalized, enterprise-first counterpoint, and a much stronger case for the staging layer than Kimball bothers to make.
- **Kimball Group** — "The 34 Subsystems of ETL".
- **Microsoft Learn** — "Best practices when working with Power Query".

One sentence to leave with. Cleaning a file is one job and building the thing that cleans it is another, so the deliverable was never the table: it is the seventeen steps, in order, with their names on them — because those are what still work tomorrow night at two in the morning when nobody is watching.